BONE PATHOLOGY II

BENIGN DISORDERS OF BONE

I. PAGET'S DISEASES

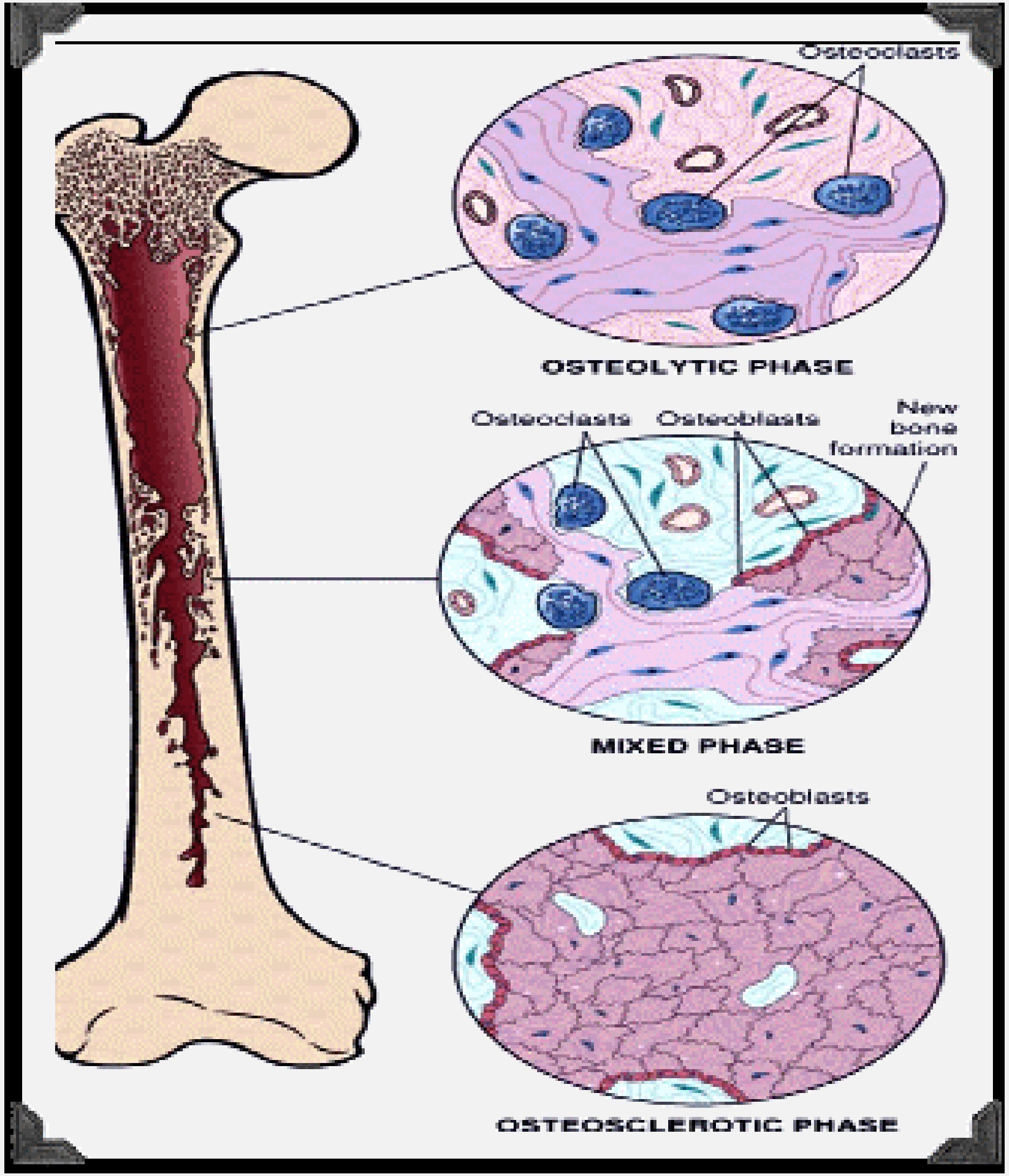
- A localized **(focal)** disorder of bone remodeling
- Second most common bone **remodeling disease** after osteoporosis.
- **Imbalance** between osteoclast and osteoblast
- More common in **whites older than 55**
- May be due to **slow virus** infection of **paramyxovirus**,

PHASES OF PAGET'S DISEASE

- **Three phases:**
  - **Osteolytic phase**: furious osteo**clastic activity** bone resorption

  Mixed phase: **osteoblast** bone formation of poorly mineralized woven bone

  Osteosclerotic phase: dense cortical and trabecular bone deposition predominates, but the bone is sclerotic and poorly organized and lacks the structural integrity and strength of normal bone.



Abnormally large **osteoclasts**

osteoclasts/osteoblasts present

newly formed bone is either woven/lamellar

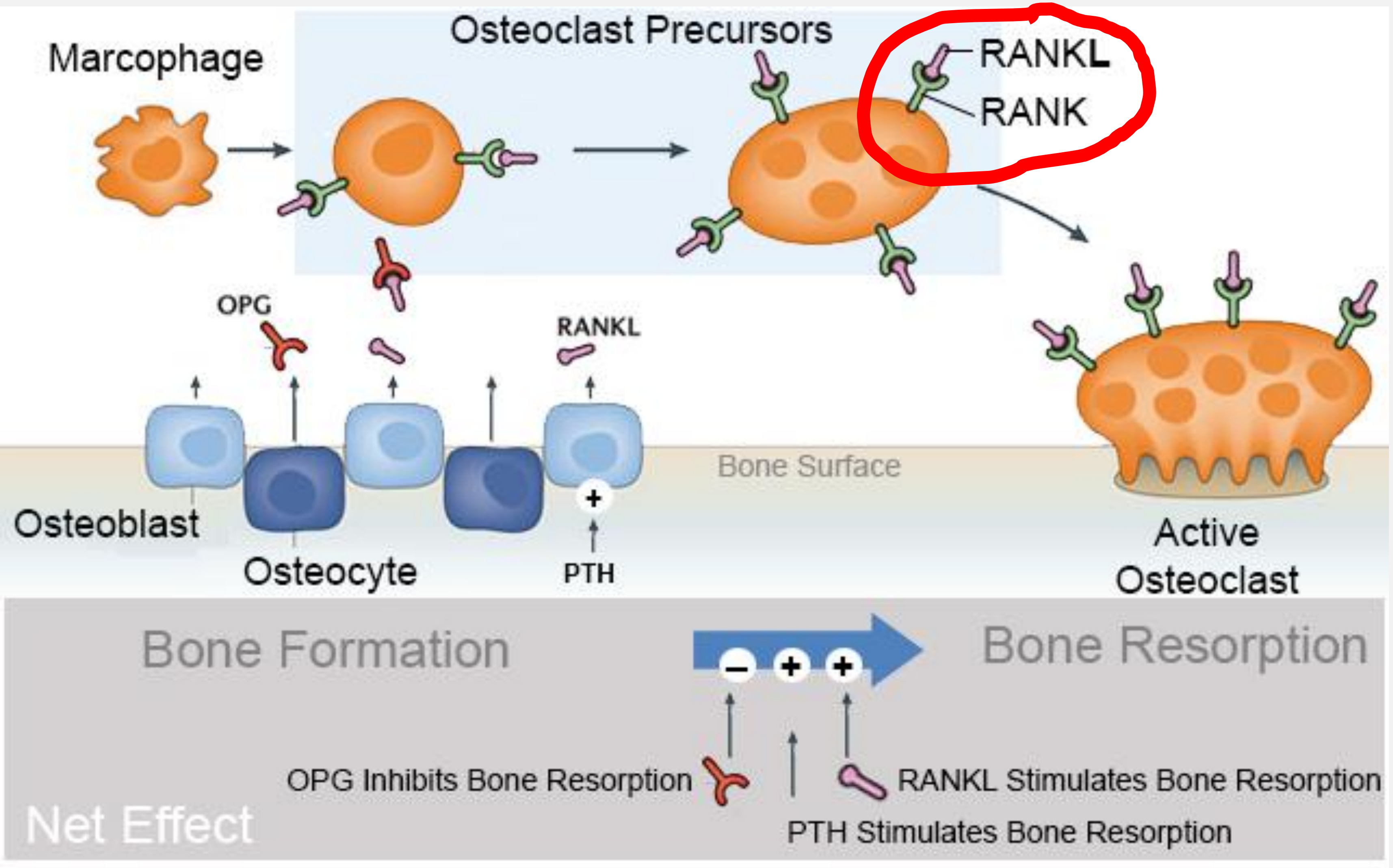
mosaic pattern
“jigsaw puzzle”

Pathogenesis of Paget's and basics in depth:

The normal adult skeleton undergoes constant **remodeling,** with osteoclasts removing bone and osteoblasts forming new bone at sites of previous bone resorption in a closely coupled fashion.

Normal remodeling of bone is under the control of **receptor activator of ligand (RANKL)** which regulates the osteoclast formation and function.

RANK present on osteoclast precursors. **RANKL, binds RANK** on osteoclast precursors promoting osteoclast proliferation and differentiation.


Marcophage
Osteoclast Precursors
RANKL
RANK
OPG
RANKL
Osteoblast
Osteocyte
PTH
Bone Surface
Active
Osteoclast
Bone Formation
Bone Resorption
OPG Inhibits Bone Resorption
RANKL Stimulates Bone Resorption
PTH Stimulates Bone Resorption
Net Effect

Pathogenesis and basics in depth:

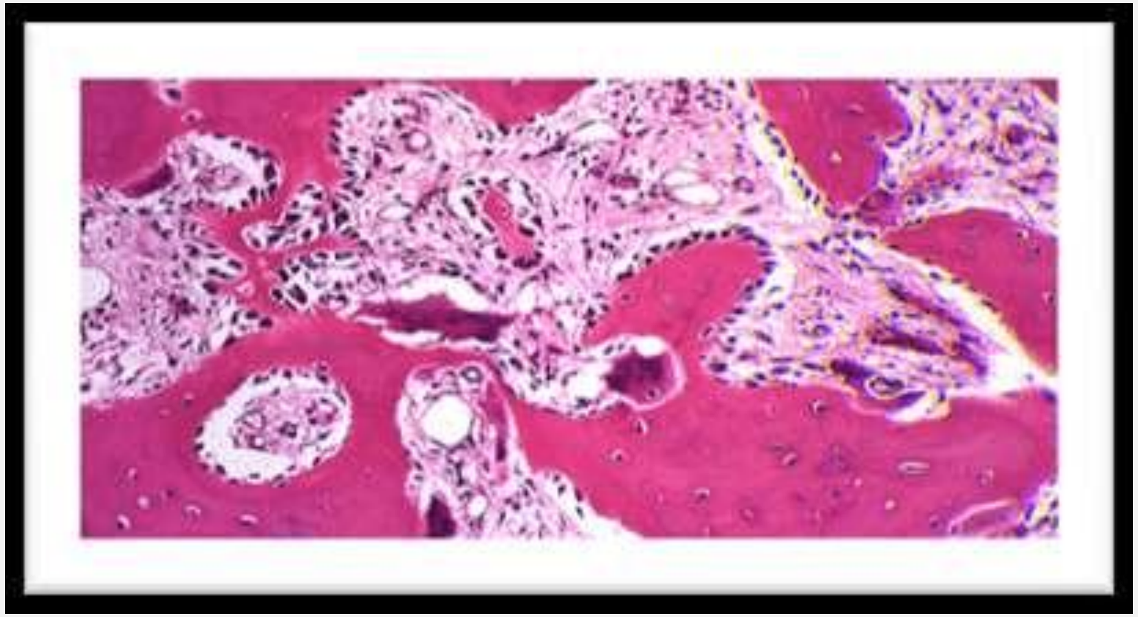
- The **osteoclast** is the primary cell affected.
- Increased hypersensitivity of osteoclast precursors to **RANKL.**

1. The initial lesion is characterized by **focal increase in osteoclastic bone resorption.**

2. followed by **accelerated bone formation :** hyperactive normal **osteoblasts**

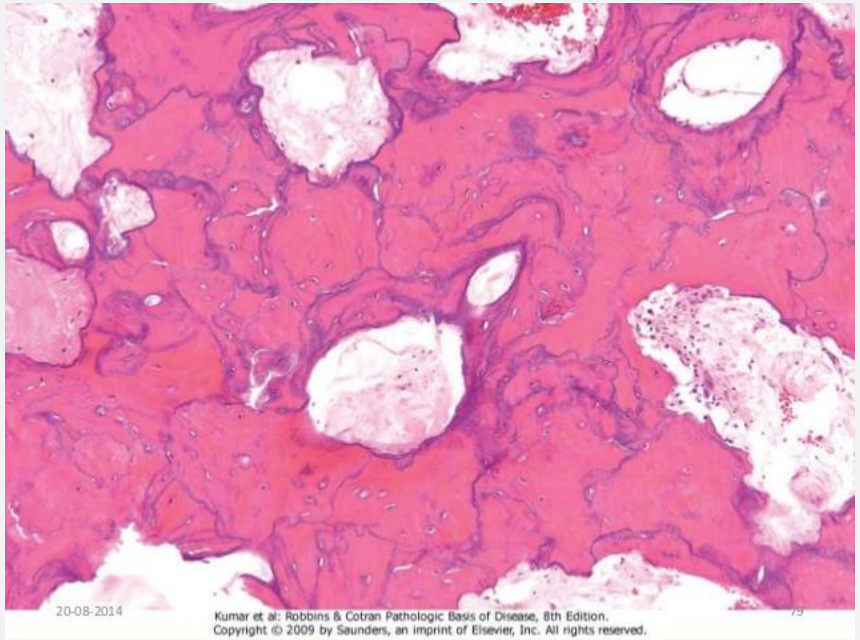
3.
- rapidly deposits new bone
- new collagen fibers are not laid down in an orderly linear fashion but rather in a disorganized manner
- a **mosaic of woven** and lamellar bone that is mechanically insufficient and at increased risk for fracture or deformity.

Mixed phase (osteolytic and osteoblastic)



giant cell and small nucleated cells

Mosaic pattern of lamellar bone in osteosclerotic stage



abnormal bone deposition

Clinical Features

- Progressive enlargement and deformity
- The most commonly involved bones include the pelvis, femur, spine, skull and tibia.
  - begins mid adulthood 50-60, progresses
  - **involves single bones**

Lab findings

- High urinary **hydroxyproline and proline:**
  - increased osteoclastic activity
  - collagen or bone matrix degradation
  - **reliable markers** of the activity of the disease.

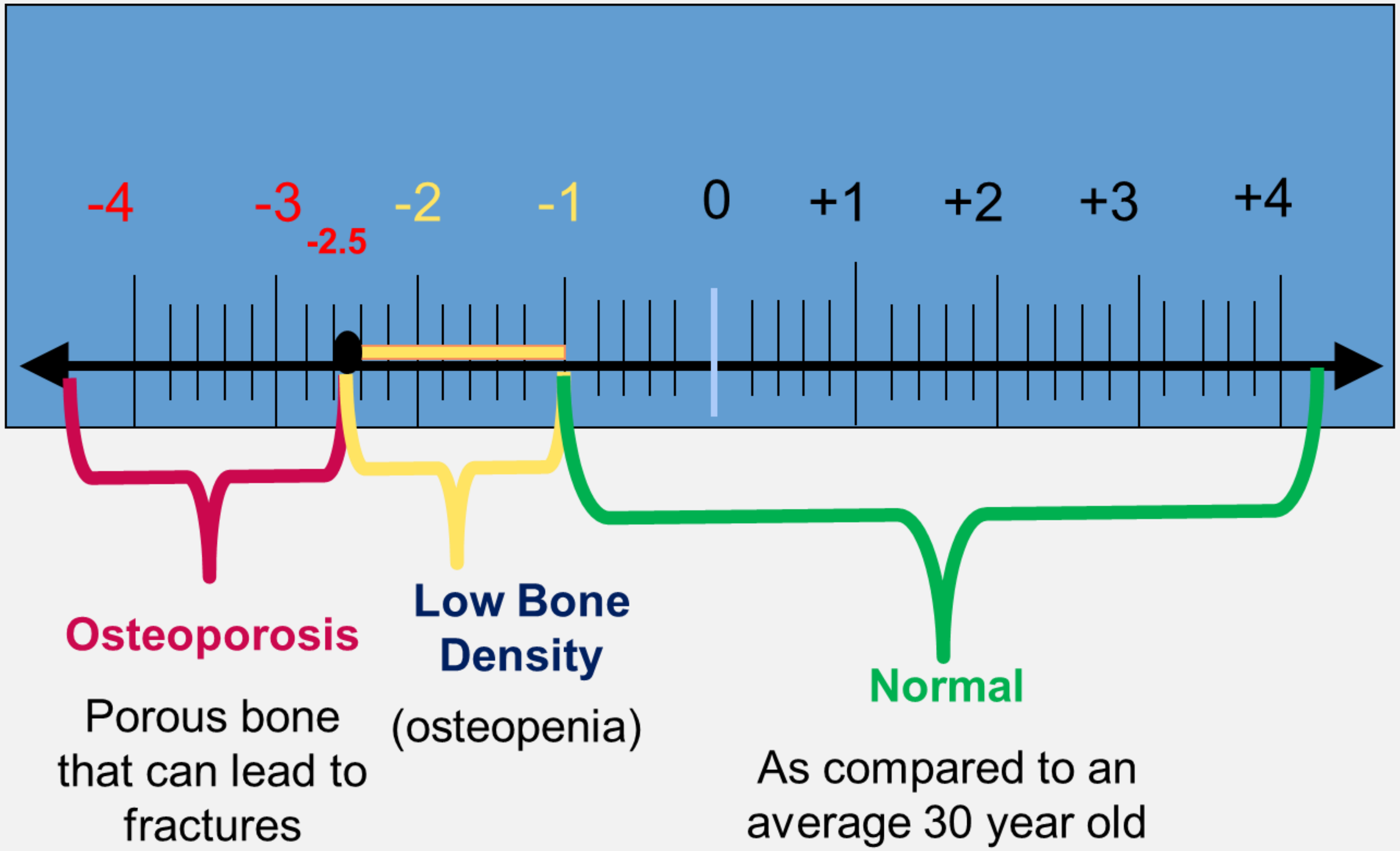
- High serum **alkaline phosphatase**:
  - Compensatory osteoblastic activity

Complications

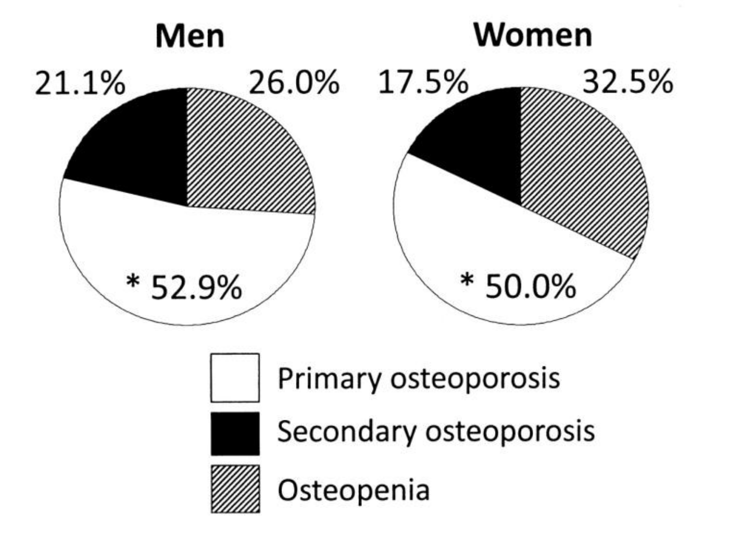
- **Sarcoma** development 5-10% precursor of osteosarcoma in elderly
- **A-V shunting** secondary to increased vascularization high output cardiac failure v

II. OSTEOPOROSIS

- Osteoporosis is defined as a systemic skeletal disease characterized by **low bone mass** and **microarchitectural** deterioration of bone tissue, with a consequent increase in bone fragility and susceptibility to **fracture**


-4
-3
-2.5
-2
-1
0
+1
+2
+3
+4
Osteoporosis
Porous bone that can lead to fractures
Low Bone Density
(osteopenia)
Normal
As compared to an average 30 year old

PRIMARY OSTEOPOROSIS	SECONDARY OTSEOPOROSIS
Aging process in conjunction with decreasing sex hormones	Medications :Anticonvulsants
	Hyperparathyroidism
	Anorexia,
	Malabsorption
	Hyperthyroidism,
	Chronic renal failure
	Cushing



PATHOPHYSIOLOGY

- imbalance of bone resorption and bone remodeling, leading to decreased skeletal mass.

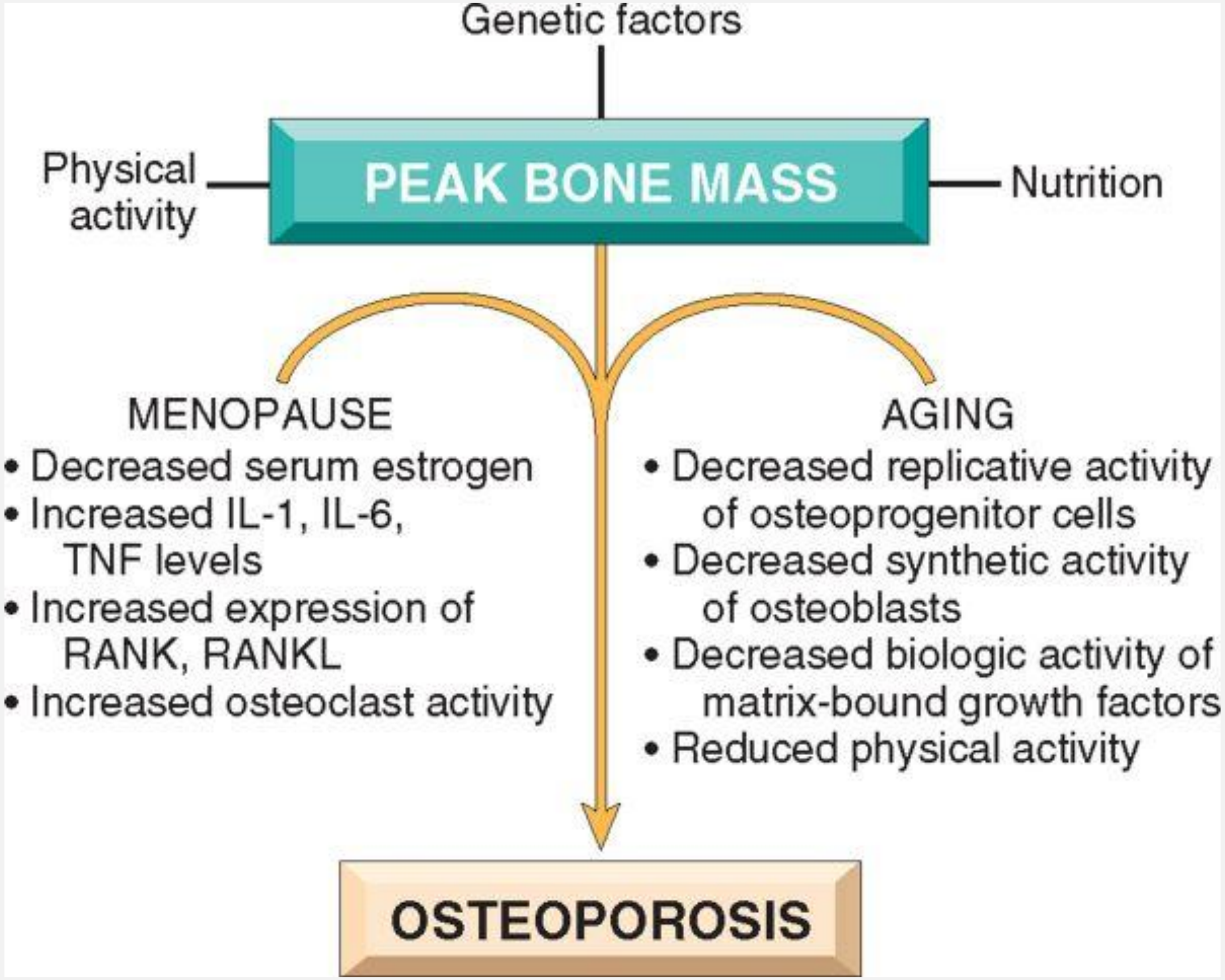
- In most individuals, **bone mass peaks** in the third decade, after which **bone resorption exceeds bone formation.**

- **Failure to reach a normal peak bone mass** or **acceleration of bone loss** can lead to osteoporosis

Peak bone mass is achieved in young adult.

Determined by :

- Hereditary account for as much as 80% of the variance in peak bone mass
- Physical activity
- Nutrition
- Hormones
- Diet



Bone loss

A small deficit in bone formation occurs every bone cycle
Normal Age-related **bone loss 0.7% every year**

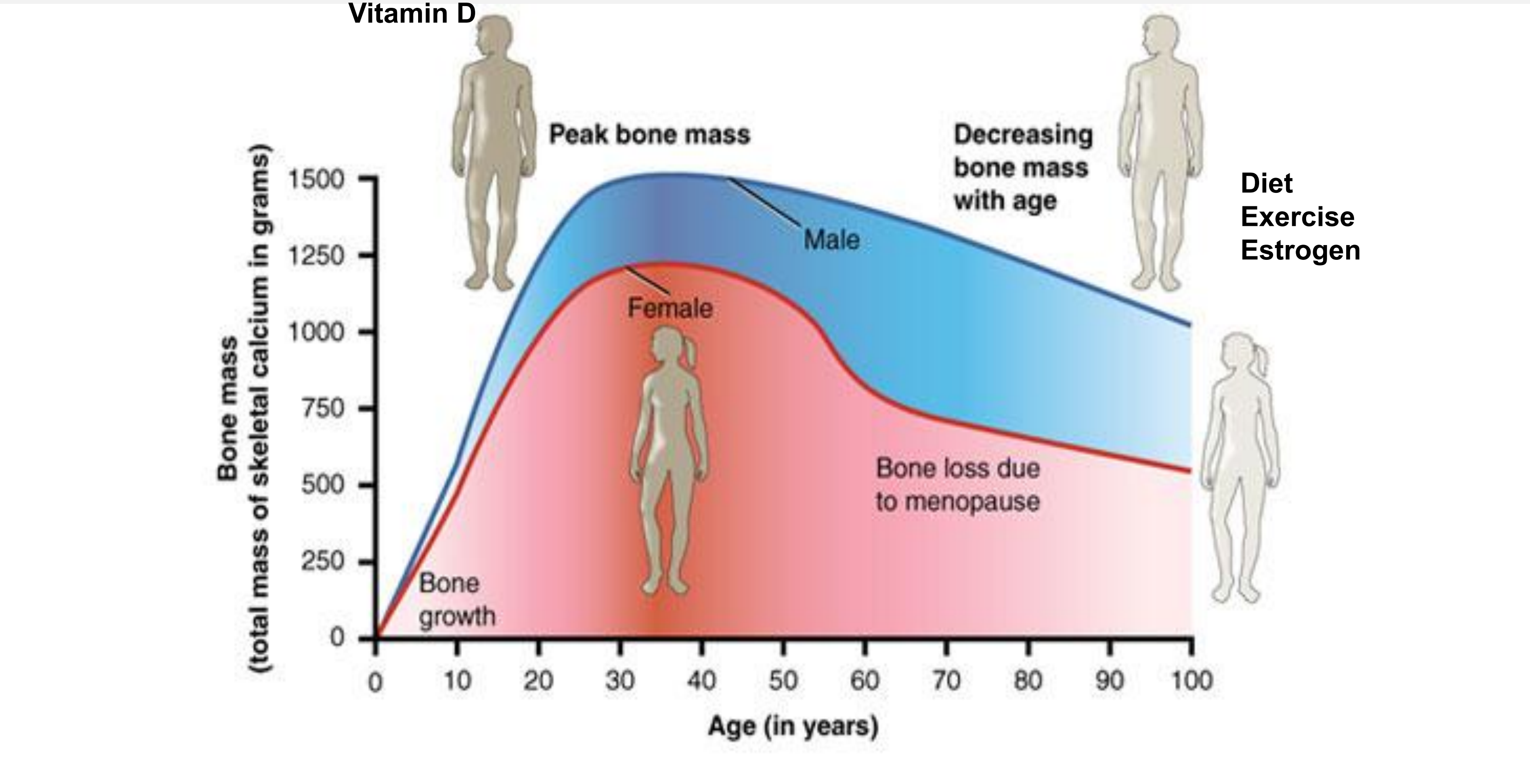
Involutional bone loss

- starts between the **ages of 35 and 45 in both sexes,**
- accelerated in the decade after the menopause in women.
- Bone loss then continues until the end of life in men and women.

Age related bone loss influenced by

- low body mass index,
- smoking,
- alcohol consumption,
- physical inactivity,
- impaired vitamin D production
- metabolism,
- secondary hyperparathyroidism.

Diet
Exercise
Vitamin D



Diet
Exercise
Estrogen

Investigation of Secondary Osteoporosis

A comprehensive history and physical includes eliciting **potential risk factors attributable to secondary bone loss.**

Investigation	Finding	Possible cause
Full blood count	Anaemia Macrocytosis	Neoplasia or malabsorption Alcohol abuse or malabsorption
Erythrocyte sedimentation rate	Raised erythrocyte sedimentation rate	Neoplasia
Biochemical profile	Hypercalcaemia Abnormal liver function Persistently high alkaline phosphatase	Hyperparathyroidism or neoplasia Alcohol abuse or liver disease Skeletal metastases or hyperparathyroidism
Thyroid function tests	Suppressed thyroid stimulating hormone; high thyroxine or triiodothyronine	Hyperthyroidism
Serum and urine immunoelectrophoresis (vertebral fractures)	Paraprotein band	Myeloma
Testosterone, sex hormone binding globulin, luteinising hormone, follicle stimulating hormone (in men)	Low testosterone or free testosterone index with abnormal gonadotrophins	Hypogonadism
Prostate specific antigen (in men)	Markedly raised levels	Skeletal metastases from prostate cancer

Morphology

Entire skeleton involved

Gross: Bone is **thin**

Bone pain and **fractures weight bearing bones**

- **Vertebrae**
- **Femoral neck**
- **Distal radius**

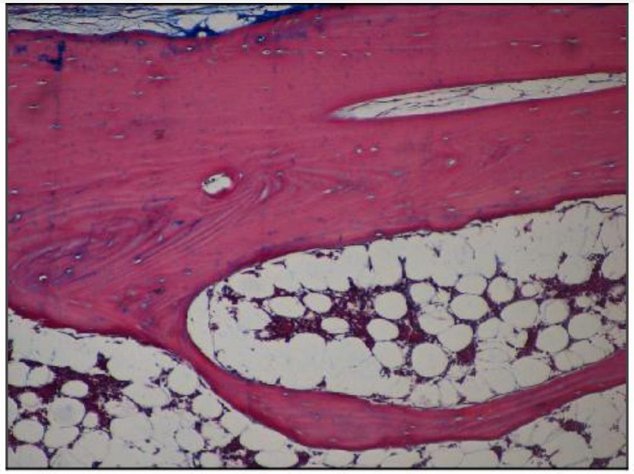
Morphology

Histology:

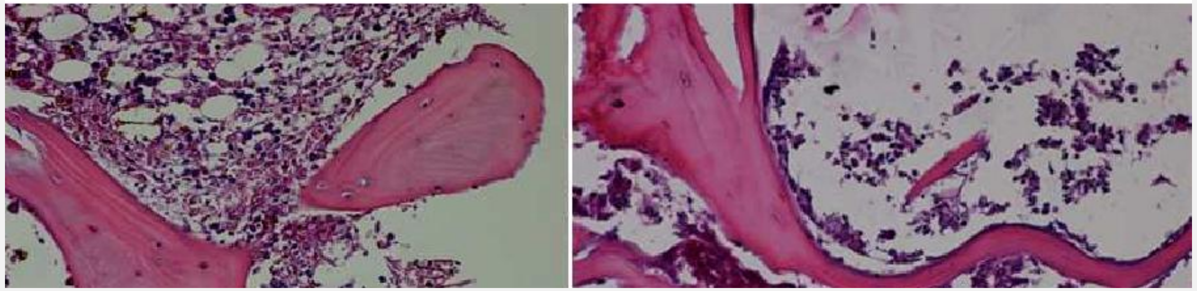
Thinned trabeculae

Decreased osteon size

Haversian canal and marrow spaces.



. 3. Thinned bone trabecula under osteoporotic conditions (Masson trichrome staining, x200).



abnormal balance between building and breaking down bone